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PREDICT STUDENT DROPOUT USING SUPERVISED AND DEEP LEARNING

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1. OVERVIEW

1.1. BUSINESS CONTEXT AND APPROACH

This report presents the findings of our study at predicting student dropout rates among adult learners. By leveraging advanced machine learning techniques, we have developed predictive models that can help identify students at risk of dropping out, enabling proactive interventions to improve student retention and educational outcomes.

1.2. DATA PREPROCESSING STRATEGY

Analysis of Study Group's dataset (25,060 records) led to a cleaned dataset of 22,680 students after removing missing values and duplicates.

The final dataset included:

- Academic: CreditWeightedAverage
- Attendance: UnauthorisedAbsenceCount, AttendancePercentage, ContactHours
- Demographics: Gender, Age
- Institutional: CentreName, CourseLevel, IsFirstIntake, ProgressionUniversity

Preprocessing included university name standardisation, categorical encoding, and numerical scaling. We addressed the target class imbalance (*See Figure A.1*) through SMOTE resampling and cross-validation.

1.3. MODEL DEVELOPMENT METHODOLOGY

We implemented a three-phase approach for XGBoost and Neural Network models:

- Established baseline models with default parameters
- Optimised through grid search cross-validation, tuning key parameters (learning rate, max depth, neurons, activation functions)
- Enhanced models by incorporating attendance features (ContactHours, AttendancePercentage)

All phases utilised 80-20 train-test splits with SMOTE balancing, evaluating performance through balanced accuracy, AUC-ROC, precision, and recall metrics.

2. ANALYSIS

2.1. EXPLORATORY DATA ANALYSIS

CreditWeightedAverage showed a clear bimodal distribution ([See Figure A.2](#)), with completing students centred around 70-80% while dropouts clustered below 20%, suggesting early academic performance strongly indicates completion risk.

Attendance patterns ([See Figure A.3](#)) proved highly discriminative: completing students maintained consistently higher attendance rates (median ~90%) compared to dropouts (median ~45%). ContactHours analysis ([See Figure A.8](#)) revealed completing students engaged with more instructional time (median ~400 hours vs ~250 hours for dropouts), while UnauthorisedAbsenceCount showed a strong negative correlation with completion ([See Figure A.9](#)), with dropouts averaging significantly more unauthorised absences.

Institutional factors showed varying impacts, with completion rates ranging from 85.4% to 100% across study centres ([See Figure A.4](#)). Pre-Masters programs ([See Figure A.10](#)) demonstrated the highest success rate (98.8%), while gender analysis ([See Figure A.6](#)) revealed higher completion rates among female students (97.1% vs 91.4% for males). These insights directly informed our feature engineering and modelling strategies.

2.2. FEATURE ENGINEERING PROCESS

Our feature engineering pipeline included:

- Age calculation from DateofBirth using a custom transformer
- Strategic encoding: one-hot for categorical variables (CentreName, Gender, ProgressionUniversity) and ordinal for CourseLevel ('Foundation' to 'Pre-Masters')
- StandardScaler normalisation for numeric features including CreditWeightedAverage and attendance metrics

2.3.MODEL IMPLEMENTATION STEPS

The dataset was split 80-20 for training and testing, with an additional 20% validation split from training data. To address the target class imbalance, SMOTE resampling was applied exclusively to the training set. Both XGBoost and Neural Network models followed a three-phase implementation:

1. Base models established performance benchmarks
2. Grid search cross-validation optimised hyperparameters (learning rate, max depth, neurons, activation functions)
3. Extended models incorporated additional attendance features

Model monitoring utilised early stopping and checkpointing to prevent overfitting, while performance evaluation focused on balanced accuracy, AUC-ROC, precision, and recall metrics to ensure robust assessment across both classes.

3. KEY INSIGHTS

3.1.MODEL PERFORMANCE COMPARISON

Table 1: XGBoost Final Metrics

Metrics Comparison:

	AUC-ROC	F1	Recall	TNR	Precision	Balanced Accuracy
XGBoost Base	0.9749	0.9771	0.9616	0.9034	0.9932	0.9325
XGBoost Tuned	0.9757	0.9743	0.9545	0.9276	0.9948	0.9411
XGBoost +Features	0.9863	0.9774	0.9592	0.9455	0.9963	0.9523

Table 2: Neural Network Final Metrics

Metrics Comparison:

	AUC-ROC	F1	Recall	TNR	Precision	Balanced Accuracy
NN Base	0.9781	0.9693	0.9435	0.9517	0.9965	0.9476
NN Tuned	0.9787	0.9710	0.9472	0.9448	0.9960	0.9460
NN Extended	0.9835	0.9698	0.9442	0.9527	0.9968	0.9484

The XGBoost model with attendance features achieved an AUC-ROC of 0.9863, slightly outperforming its Neural Network counterpart (0.9835). This marginal difference suggests both approaches are viable for dropout prediction.

Key performance insights:

- ✓ XGBoost showed superior balanced accuracy (0.9523) compared to NN (0.9484) when including attendance features, indicating better handling of class imbalance.
- ✓ Neural Network demonstrated stronger True Negative Rate (0.9527 vs 0.9455), suggesting better reliability in identifying students likely to complete.
- ✓ Both models achieved remarkably high precision (XGBoost: 0.9963, NN: 0.9968), indicating extremely reliable positive predictions.
- ✓ The addition of attendance features improved all key metrics across both models, with XGBoost showing more substantial gains in balanced accuracy compared to NN.

3.2.FEATURE IMPORTANCE ANALYSIS

CreditWeightedAverage emerged as the dominant predictor across all XGBoost models (*See Figure B.4*), accounting for approximately 51-57% of predictive power. This overwhelming influence suggests academic performance serves as the primary early warning indicator for dropout risk.

Different International Study Centres showed varying levels of influence:

- ISC Kingston and Sheffield consistently ranked among top predictors (3-4% importance each)
- Center-specific variations suggest institutional factors play a significant role in student retention

Gender and intake timing (IsFirstIntake) showed modest but consistent influence (2-3% importance), suggesting these characteristics provide valuable secondary signals for dropout prediction.

3.3.IMPACT OF ATTENDANCE FEATURES

When included, AttendancePercentage (3.15%) and ContactHours (2.53%) ranked as 5th and 7th most important features respectively (*See Figure B.4*), validating their inclusion for improved model performance.

4. BEST PERFORMING METHOD

4.1.MODEL SELECTION RATIONALE

XGBoost with attendance features emerged as the optimal solution, outperforming Neural Networks with superior balanced accuracy (0.9523 vs 0.9484) and AUC-ROC (0.9863 vs 0.9835). While both models demonstrated exceptional predictive capability, XGBoost offers key advantages: clear feature importance rankings for targeted interventions, faster training time for efficient updates, and robust handling of class imbalance (Precision: 0.9963, Recall: 0.9592) to minimise both false alarms and missed identifications.

4.2.IMPLEMENTATION CONSIDERATIONS

- Deploy real-time monitoring of CreditWeightedAverage and attendance metrics
- Establish centre-specific early warning thresholds
- Implement automated alerts for at-risk students
- Develop targeted first-intake support programs

5. VISUALISATION ANALYSIS

5.1.ROC CURVES AND CONFUSION MATRICES

ROC curve analysis (*See Figure B.1*) demonstrates progressive improvement across model iterations. The steeper initial curve indicates excellent sensitivity in identifying high-risk students. Confusion matrices (*See Figure B.2*) reveal evolving prediction patterns:

- XGBoost+Features achieved the lowest false positives (15) compared to base (28) and tuned (21) versions

- Neural Network extended model similarly reduced false positives (13) while maintaining high true positive rates
- Both final models maximised true positive predictions while minimising false alarms

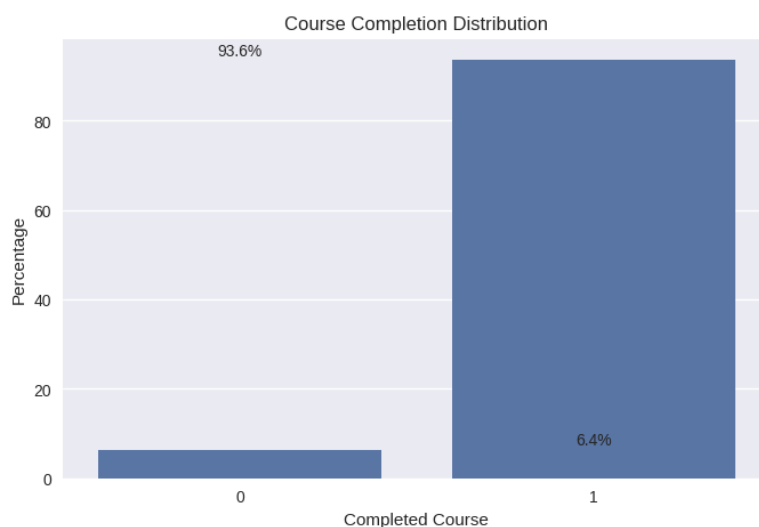
5.2.LEARNING CURVES AND MODEL TRAINING

Neural Network variants exhibited distinct training patterns (*See Figure B.3*):

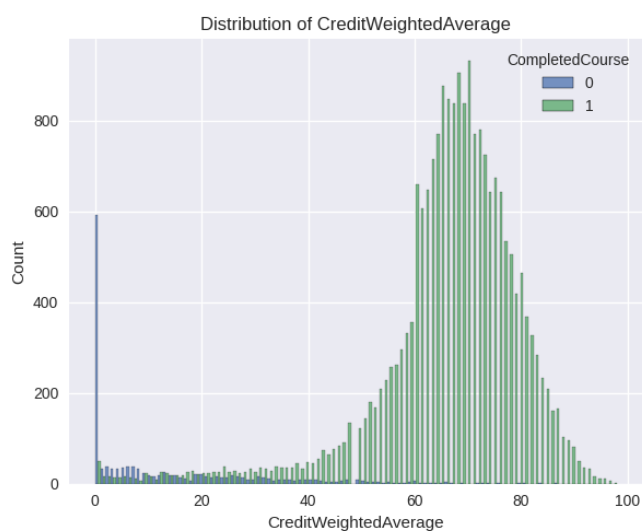
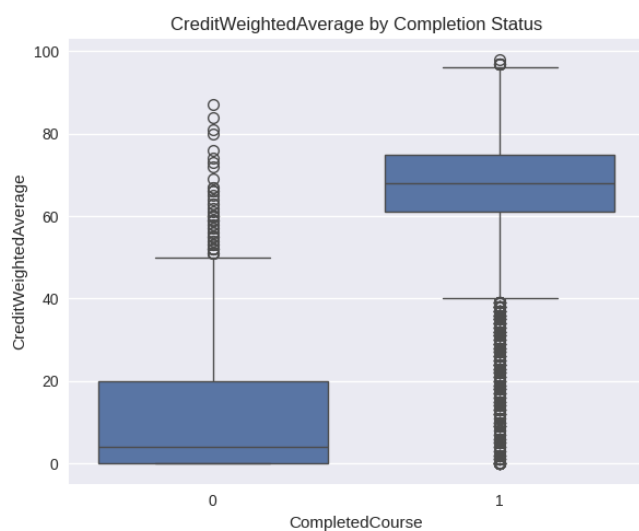
- Base model showed rapid initial convergence with moderate validation loss fluctuation.
- Tuned model achieved lower initial loss (0.26) with improved stability.
- Extended model, despite higher initial loss (1.2), demonstrated optimal convergence over 25 epochs with minimal training-validation gap, indicating robust generalisation.

6. APPENDIX A: EDA VISUALISATIONS

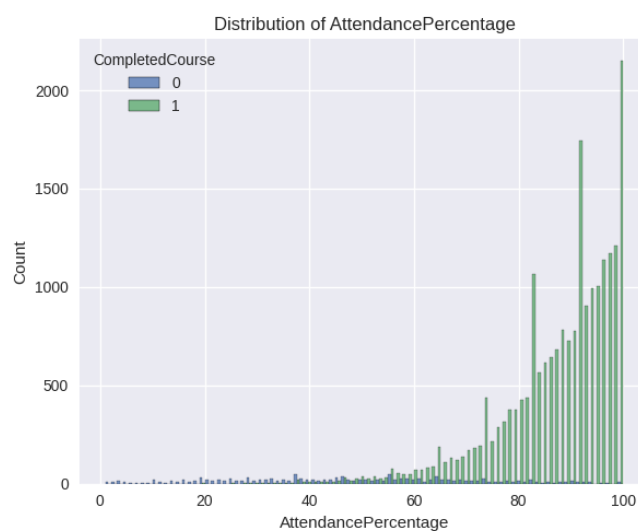
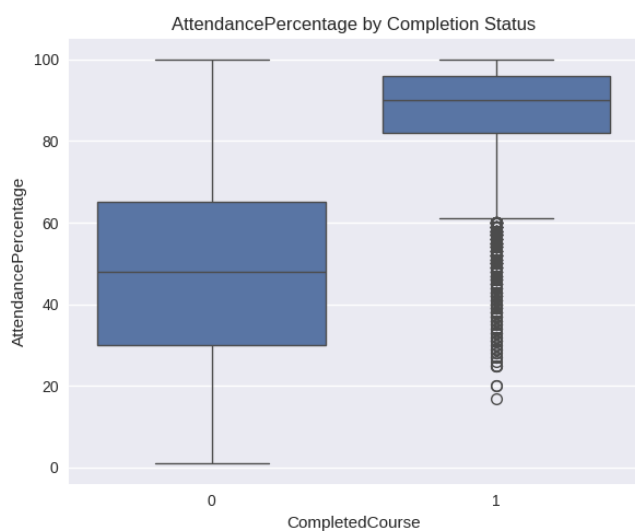
A.1: Course Completion Distribution and Rates



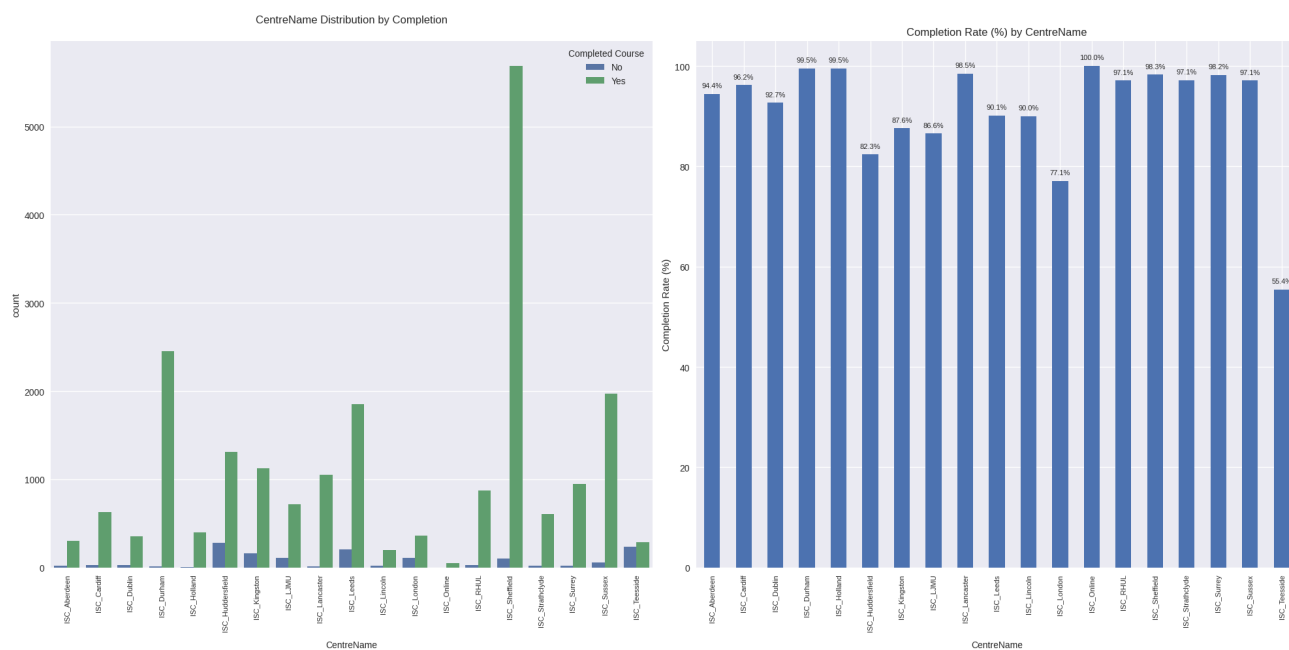
A.2: Credit Weighted Average Analysis



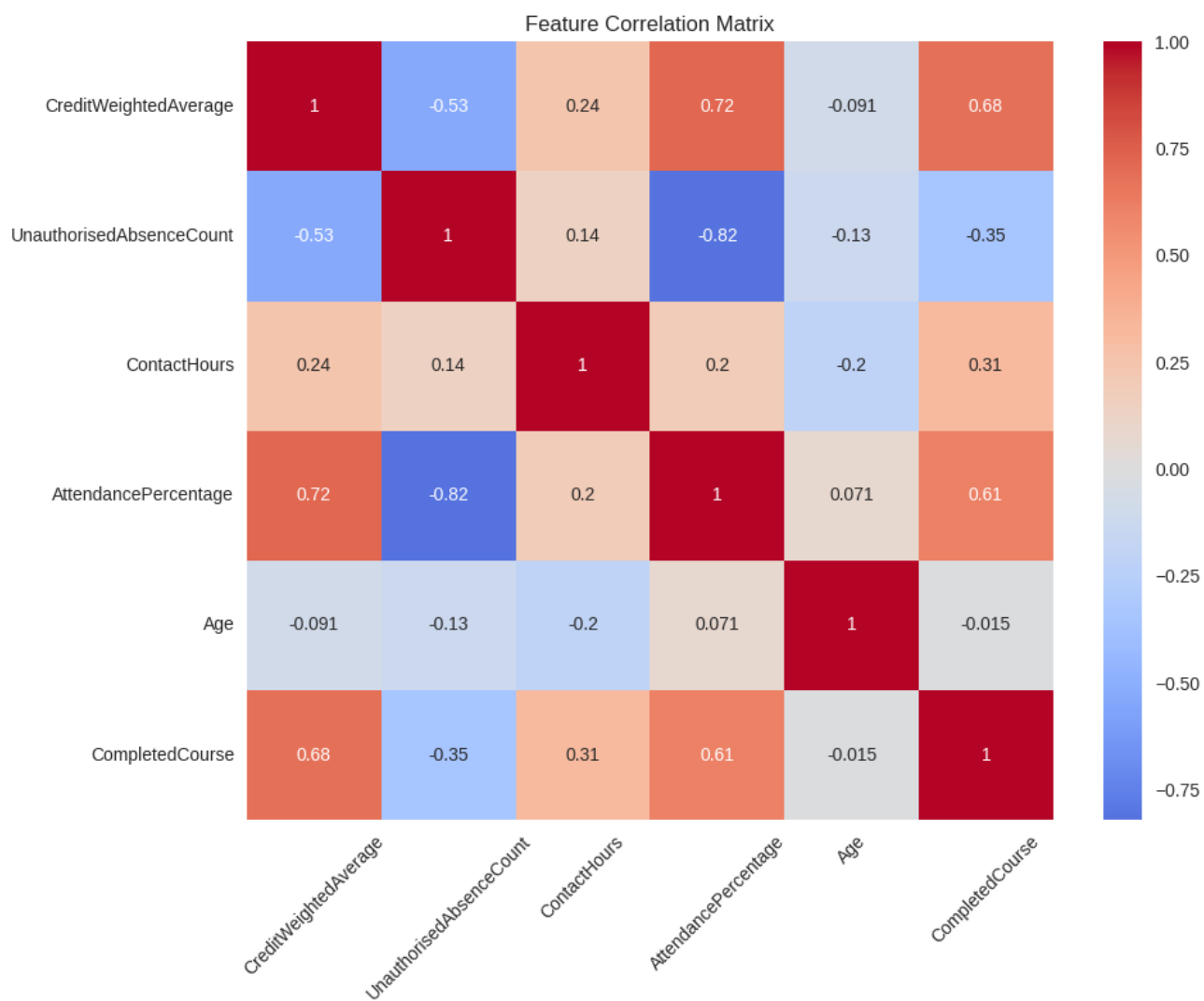
A.3: Attendance Metrics Analysis



A.4: Study Center Performance

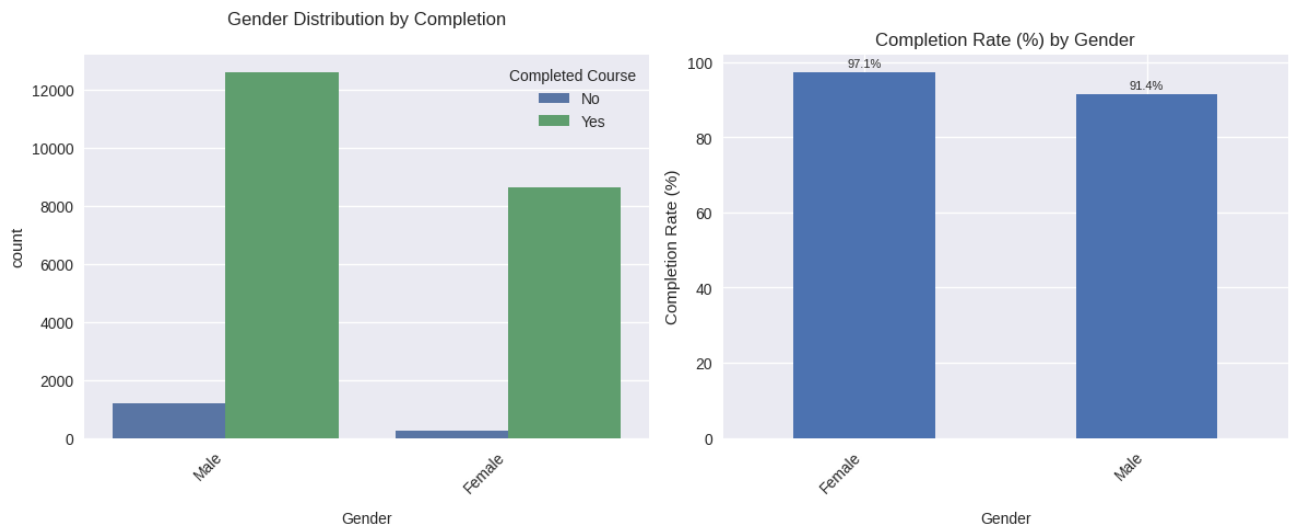


A.5: Feature Correlation Matrix

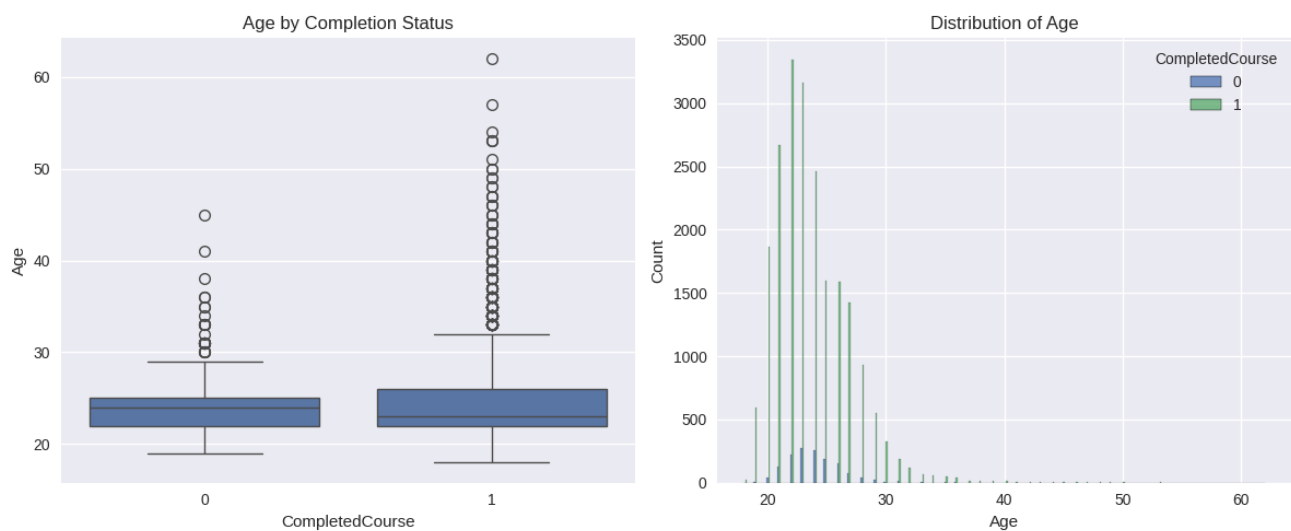


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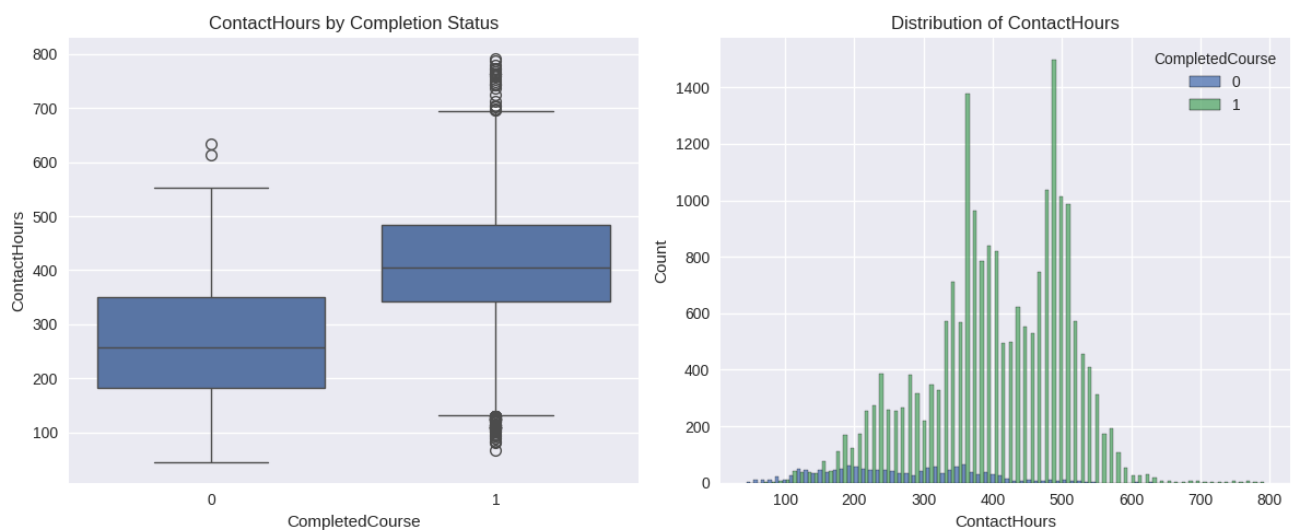
A.6: Gender Analysis



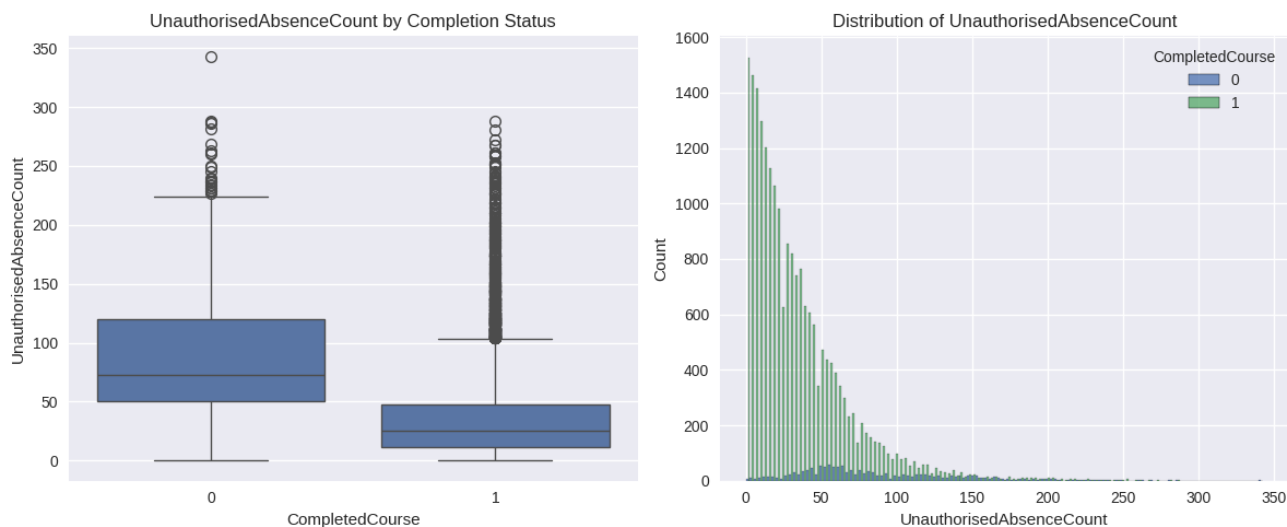
A.7: Age Analysis



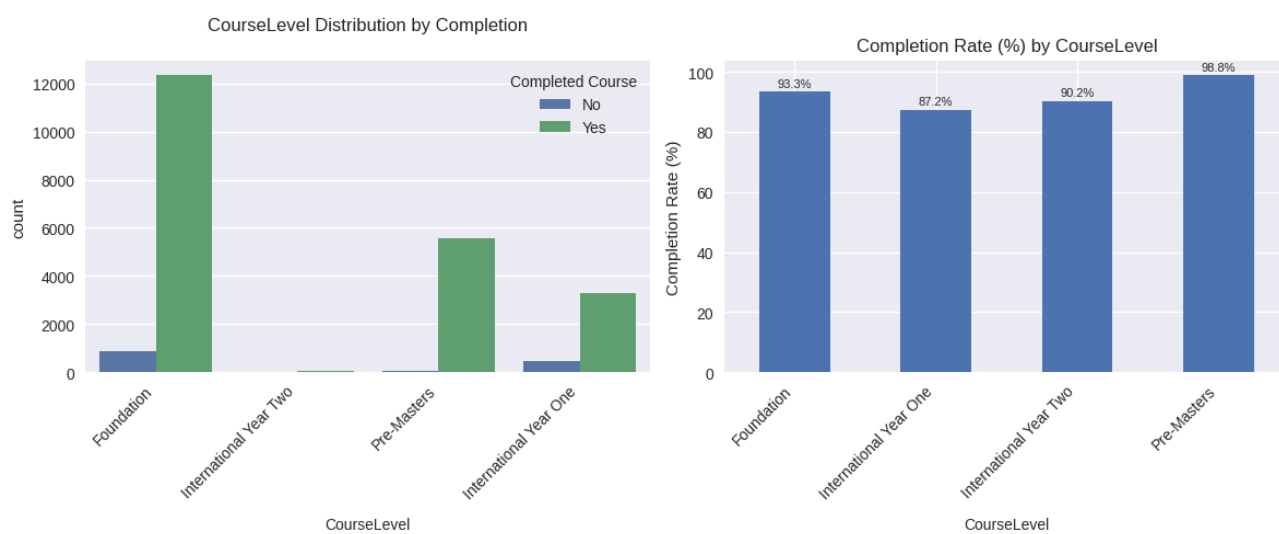
A.8: ContactHours Distribution



A.9: UnauthorisedAbsenceCount Distribution

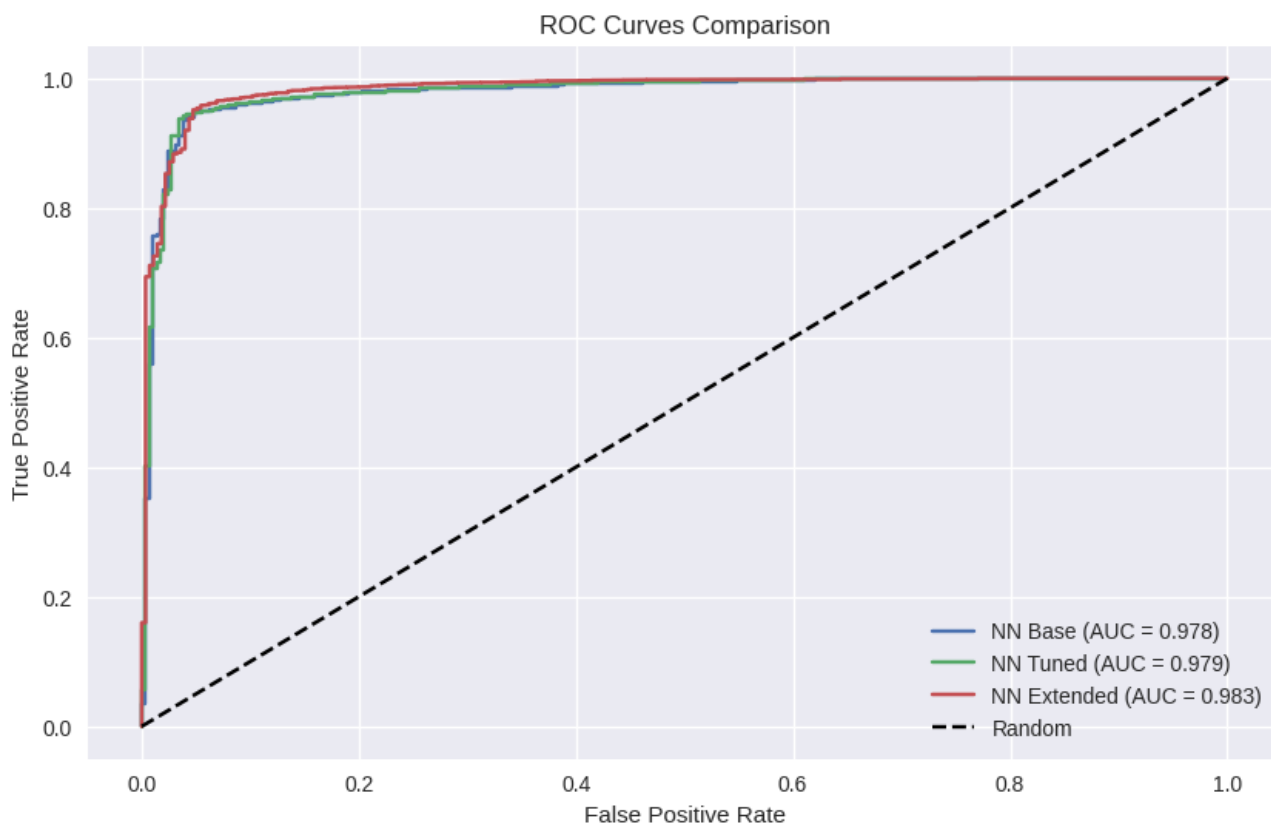
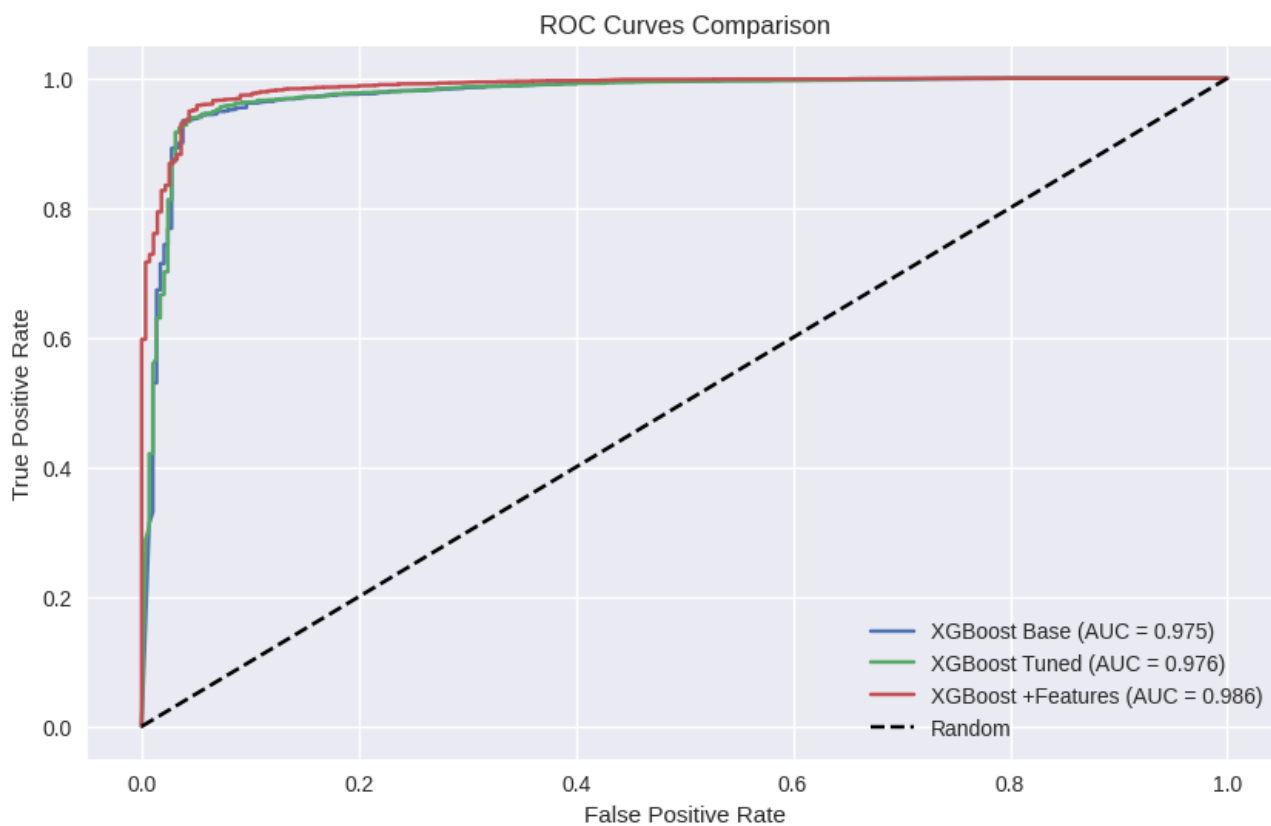


A.10: CourseLevel Distribution

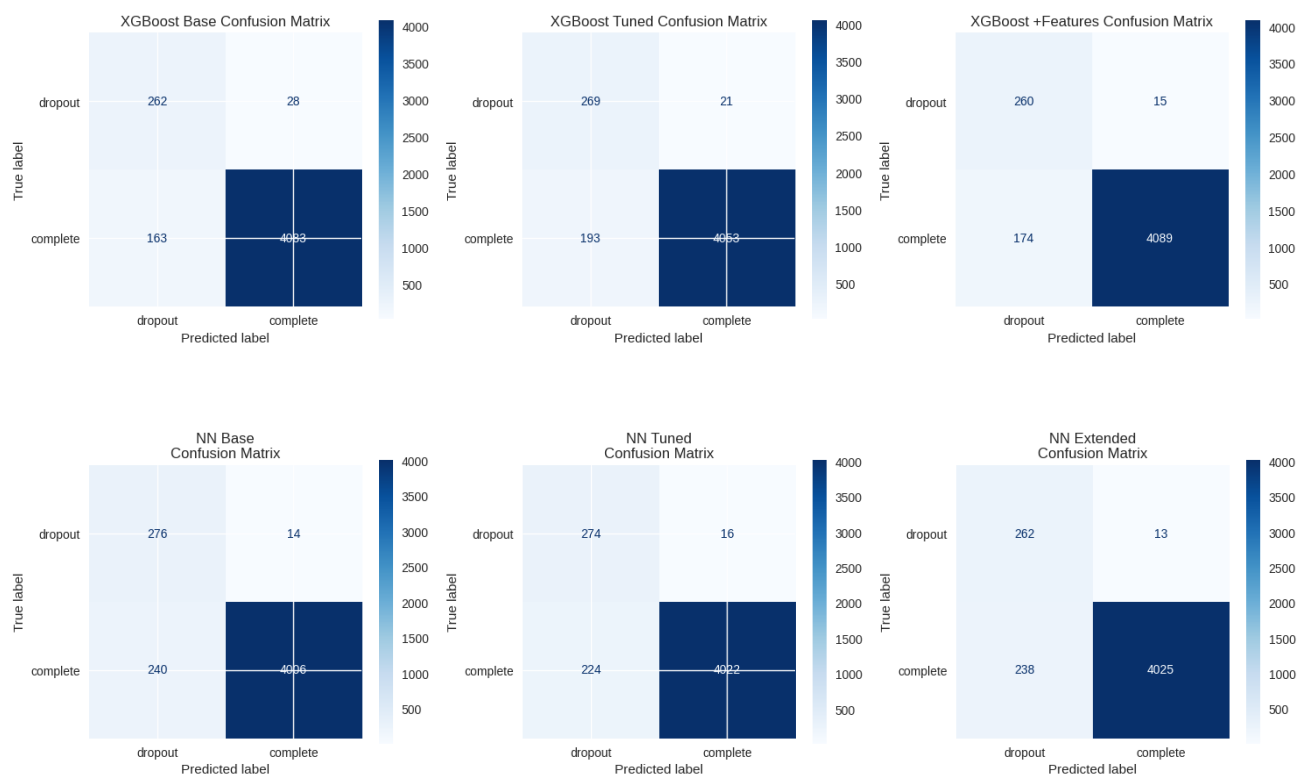


7. APPENDIX B: MODEL PERFORMANCE VISUALISATIONS

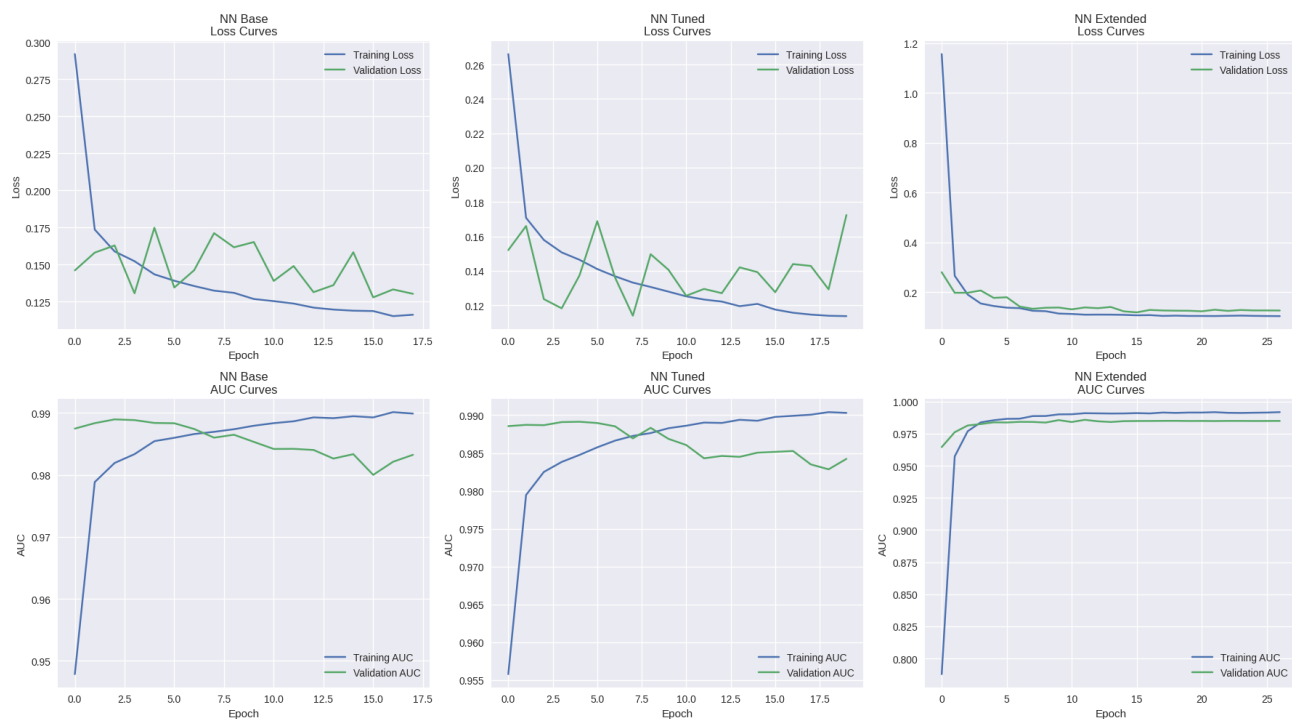
B.1: ROC Curves Comparison



B.2: Confusion Matrices



B.3: Neural Network Learning Curves



B.4: Feature Importance Rankings

